

Existence and Uniqueness for Periodic KdV in H^3

Problem Setup

KdV on $(-L, L)$ with periodic BC:

$$u_t + 6uu_x + u_{xxx} = 0$$

Initial data:

$$u(x, 0) = u_0(x), \quad u_0 \in H_{\text{per}}^3(-L, L)$$

Goal: Prove existence and uniqueness in H^3 .

Strategy

We use:

- ▶ Energy estimates in H^3
- ▶ Sobolev embedding in 1D
- ▶ Gronwall inequality

Gronwall's Inequality

Lemma (Gronwall)

Let $y(t) \geq 0$ satisfy:

$$\frac{d}{dt}y(t) \leq C y(t), \quad y(0) = y_0$$

Then: $y(t) \leq y_0 e^{Ct}$

Proof: Rewrite:

$$\frac{y'(t)}{y(t)} \leq C$$

Integrate from 0 to t :

$$\int_0^t \frac{y'(s)}{y(s)} ds \leq Ct$$

$$\ln y(t) - \ln y_0 \leq Ct$$

Exponentiate:

$$y(t) \leq y_0 e^{Ct}$$

Finite-Time Blow-Up Inequality

Lemma

Let $y(t) \geq 0$ satisfy $y'(t) \leq C y(t)^2$, with $y(0) = y_0 > 0$
Then $y(t) \leq \frac{y_0}{1 - C t y_0}$ for $t < \frac{1}{C y_0}$.

Proof:

$$\frac{y'(t)}{y(t)^2} \leq C \quad \implies \quad \frac{d}{dt} \left(-\frac{1}{y(t)} \right) = \frac{y'(t)}{y(t)^2} \leq C$$

Integrating from 0 to t gives $-\frac{1}{y(t)} + \frac{1}{y_0} \leq C t$

$$\frac{1}{y(t)} \geq \frac{1}{y_0} - C t \quad \implies \quad y(t) \leq \frac{y_0}{1 - C t y_0}$$

Interpretation: Blow up no earlier than $t = \frac{1}{C y_0}$.

What Does an Embedding Mean?

Idea: An embedding says that one function space is *contained in* another, with control of norms.

Definition (informal):

$$X \hookrightarrow Y \quad \text{means} \quad X \subset Y \text{ and } \|u\|_Y \leq C\|u\|_X$$

Interpretation:

- ▶ Functions in X are automatically more regular than expected
- ▶ The X norm controls the Y norm
- ▶ This is a *quantitative inclusion*

Example (1D):

$$H^1(-L, L) \hookrightarrow L^\infty(-L, L)$$

Meaning:

- ▶ Square-integrable derivative $\Rightarrow$ bounded function
- ▶ Oscillations are controlled by energy

Sobolev Embedding (1D)

Lemma

For periodic functions:

$$H^1(-L, L) \hookrightarrow L^\infty(-L, L)$$

Thus:

$$\|u\|_{L^\infty} \leq C \|u\|_{H^1}$$

In particular:

$$H^3 \subset C^2$$

Proof of 1D Sobolev Embedding

Proof:

Fix $x \in (-L, L)$ and write:

$$u(x) = \frac{1}{2L} \int_{-L}^L u(y) dy + \int_{-L}^x u'(s) ds - \frac{1}{2L} \int_{-L}^L \int_{-L}^y u'(s) ds dy$$

Estimate each term:

$$\left| \frac{1}{2L} \int_{-L}^L u \right| \leq \frac{1}{\sqrt{2L}} \|u\|_{L^2}$$

$$\left| \int_{-L}^x u'(s) ds \right| \leq \sqrt{2L} \|u'\|_{L^2}$$

Thus:

$$|u(x)| \leq C(\|u\|_{L^2} + \|u'\|_{L^2}) = C\|u\|_{H^1}$$

Taking supremum over x gives the result.

Proof of 1D Sobolev Embedding (Simple Version)

Lemma

For $u \in H^1(-L, L)$, $\|u\|_{L^\infty(-L, L)} \leq C \|u\|_{H^1(-L, L)}$

Proof:

Fix $x \in (-L, L)$. Using the Fundamental Theorem of Calculus:

$$u(x) = u(-L) + \int_{-L}^x u'(s) ds$$

Estimate:

$$|u(x)| \leq |u(-L)| + \int_{-L}^L |u'(s)| ds$$

Use Cauchy-Schwarz:

$$\int_{-L}^L |u'(s)| ds \leq \sqrt{2L} \|u'\|_{L^2}$$

All we need to do now is justify

$$|u(-L)| \leq C \|u\|_{H^1}$$

Product Estimate in H^3

Lemma

If $u \in H^3$, then:

$$\|uu_x\|_{H^3} \leq C\|u\|_{H^3}^2$$

Idea: Differentiate up to order 3 and use:

$$\|fg\|_{L^2} \leq \|f\|_{L^\infty}\|g\|_{L^2}$$

All terms controlled since $H^2 \subset L^\infty$.

L^2 Energy Estimate

Multiply equation by u :

$$\frac{d}{dt} \int u^2 dx = 0$$

Thus:

$$\|u(t)\|_{L^2} = \|u_0\|_{L^2}$$

H^1 Estimate

Differentiate once:

$$(u_x)_t + 6(uu_x)_x + u_{xxxx} = 0$$

Multiply by u_x and integrate:

$$\frac{d}{dt} \int u_x^2 dx \leq C \|u\|_{H^3} \|u_x\|_{L^2}^2$$

H^3 Energy Estimate

Apply ∂_x^3 :

$$\partial_t u^{(3)} + 6\partial_x^3(uu_x) + u^{(6)} = 0$$

Multiply by $u^{(3)}$ and integrate:

$$\frac{d}{dt} \|u\|_{H^3}^2 \leq C \|u\|_{H^3}^3$$

Key point: highest derivatives cancel by integration by parts.

A Priori Bound

We obtain:

$$\frac{d}{dt} \|u\|_{H^3}^2 \leq C \|u\|_{H^3}^3$$

Thus:

$$\frac{d}{dt} \|u\|_{H^3} \leq C \|u\|_{H^3}^2$$

By Gronwall:

$$\|u(t)\|_{H^3} \leq \frac{\|u_0\|_{H^3}}{1 - Ct\|u_0\|_{H^3}}$$

Local Existence

Theorem

For $u_0 \in H_{per}^3$, there exists $T > 0$ and a solution:

$$u \in C([0, T]; H_{per}^3)$$

Proof via:

- ▶ Picard iteration OR Galerkin approximation
- ▶ Uniform H^3 bounds
- ▶ Compactness

The Linear Solution Operator $S(t)$

Consider the linear Airy equation:

$$u_t + u_{xxx} = 0, \quad u(x, 0) = u_0(x)$$

Definition: The linear solution operator $S(t)$ is defined by

$$S(t)u_0 = u(t)$$

where $u(t)$ solves the linear equation.

Fourier representation (periodic case):

Write

$$u_0(x) = \sum_{k \in \mathbb{Z}} \hat{u}_k e^{ik\pi x/L}$$

Then:

$$S(t)u_0 = \sum_{k \in \mathbb{Z}} e^{-i(k\pi/L)^3 t} \hat{u}_k e^{ik\pi x/L}$$

Key properties:

- ▶ $S(t)$ is a linear group: $S(t+s) = S(t)S(s)$
- ▶ $S(0) = I$
- ▶ $S(t)$ is unitary on L^2 :

Picard Iteration for KdV

We rewrite KdV:

$$u_t + u_{xxx} = -6uu_x$$

Using the linear solution operator $S(t)$:

$$u(t) = S(t)u_0 - 6 \int_0^t S(t-\tau)(uu_x)(\tau) d\tau$$

Idea: Build the solution by iteration.

Define:

$$u^{(0)}(t) = S(t)u_0$$

$$u^{(n+1)}(t) = S(t)u_0 - 6 \int_0^t S(t-\tau)(u^{(n)}u_x^{(n)})(\tau) d\tau$$

Each step solves a *linear* equation with known forcing.

Why Picard Iteration Works

Work in:

$$X_T = C([0, T]; H^3_{\text{per}})$$

Step 1: Uniform bound

Using $\|uu_x\|_{H^3} \leq C\|u\|_{H^3}^2$:

$$\|u^{(n+1)}\|_{X_T} \leq \|u_0\|_{H^3} + CT\|u^{(n)}\|_{X_T}^2$$

For T small, all iterates stay bounded.

Step 2: Contraction

Let $w^{(n)} = u^{(n)} - u^{(n-1)}$:

$$\|w^{(n+1)}\|_{X_T} \leq CTR\|w^{(n)}\|_{X_T}$$

If $CTR < 1$, iteration contracts.

Conclusion:

- ▶ $u^{(n)} \rightarrow u$ in X_T
- ▶ u solves KdV
- ▶ Uniqueness follows from same estimate

Uniqueness (Alt)

Let $w = u - v$.

Then:

$$w_t + 6(uw_x + wv_x) + w_{xxx} = 0$$

Energy estimate:

$$\frac{d}{dt} \|w\|_{L^2}^2 \leq C(\|u\|_{H^3} + \|v\|_{H^3}) \|w\|_{L^2}^2$$

By Gronwall:

$$w = 0$$

Global Existence

Use conserved quantities:

$$\int u^2 dx, \quad \int (u_x^2 - 2u^3) dx$$

These control $\|u\|_{H^1}$.

Bootstrap:

- ▶ H^1 bounded globally
- ▶ H^3 cannot blow up in finite time

Thus solution extends for all t .

Final Theorem

Theorem

If $u_0 \in H_{per}^3(-L, L)$, then KdV admits a unique global solution:

$$u \in C([0, \infty); H_{per}^3)$$

Key simplifications in H^3 :

- ▶ No delicate Sobolev product theory
- ▶ Classical energy estimates suffice
- ▶ Fully compatible with spectral methods